Functional Requirements Document – Board

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document (sub-module 02.3 Board, code `BOARD`) · **Version:** 1.0 · **Status:** Draft
> **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026
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> **Based on:** BRD-Board.md v1.0
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Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version, derived from BRD-Board v1.0 and the verified Kanban Board workflow doc

Related Documents

Document	Purpose	Location
BRD – Board	Business needs this FRD details (`REQ-BOARD-nn`, `BR-BOARD-nn`)	`../../../BRD/02-TaskFlow-Monitoring/02.3-Board/BRD-Board.md`
Overall BRD / FRD	Roles, global rules `BR-nn`, global NFRs	`../../../BRD/00-Overall/BRD-TaskFlow-Overall.md`, `../../00-Overall/FRD-TaskFlow-Overall.md`
Sibling FRDs – Task Management, Employee Workload	Task scope, status rules, history, Task Details	`../02.1-Task-Management/FRD-Task-Management.md`, `../02.2-Employee-Workload/FRD-Employee-Workload.md`
Development Plan	Screen-wise build plan (original Screen 10)	`TF - Documents/Screens Wise Feature Development Plan/10-kanban-board-screen.md`
HLD / LLD / TDD	Technical design	`../../../HLD/02-TaskFlow-Monitoring/02.3-Board/HLD-Board.md`, `../../../LLD/02-TaskFlow-Monitoring/02.3-Board/LLD-Board.md`, `TDD-TaskFlow.md` (v2.0)

1. Introduction

1.1 Purpose

This FRD specifies the detailed, testable behaviour of the Board (Kanban): which tasks appear in which column, what a card shows, how a drag changes a task's status under the shared status rules, how the filters behave, and who may use each part.

1.2 Scope

In scope: the Board screen (`/board`) – five status columns, task cards, drag-and-drop status change, Project / Assignee / Status / Deadline filters, Refresh and opening a task from a card.

Out of scope (see BRD §3.2): a Cancelled column; re-ordering cards within a column; paging; notifications for a status change; editing task details on the board (Task Details, `../02.1-Task-Management/FRD-Task-Management.md`).

1.3 Audience

Developer, tester and reviewer of TaskFlow (Jojo George), and the product owner approving the behaviour.

1.4 Conventions

- Requirement IDs: `FR-BOARD-nn`. Use cases: `UC-BOARD-nn`. BRD requirements: `REQ-BOARD-nn`; sub-module rules `BR-BOARD-nn`; Task Management rules `BR-TASK-nn`; global rules `BR-nn` (overall BRD).
- Priority: **Must / Should / Could** (MoSCoW).
- "The system shall ..." = mandatory behaviour; "should" = recommended.
- Text in quotes is the exact UI label or message; `{name}` marks a value filled in at run time.

1.5 Definitions & Abbreviations

Term	Meaning
Board / Kanban	A view of tasks as columns, one per status
Card	One task on the board
Column	A status: To Do, In Progress, In QA, Blocked or Completed
In QA	The task is being checked before it is completed
Overdue card	A card whose deadline has passed and whose task is neither Completed nor Cancelled
Task scope	The tasks a user may see: Own, Team or Organization (as on the Task List)
Optimistic move	The card moves on screen at once and moves back if the server refuses

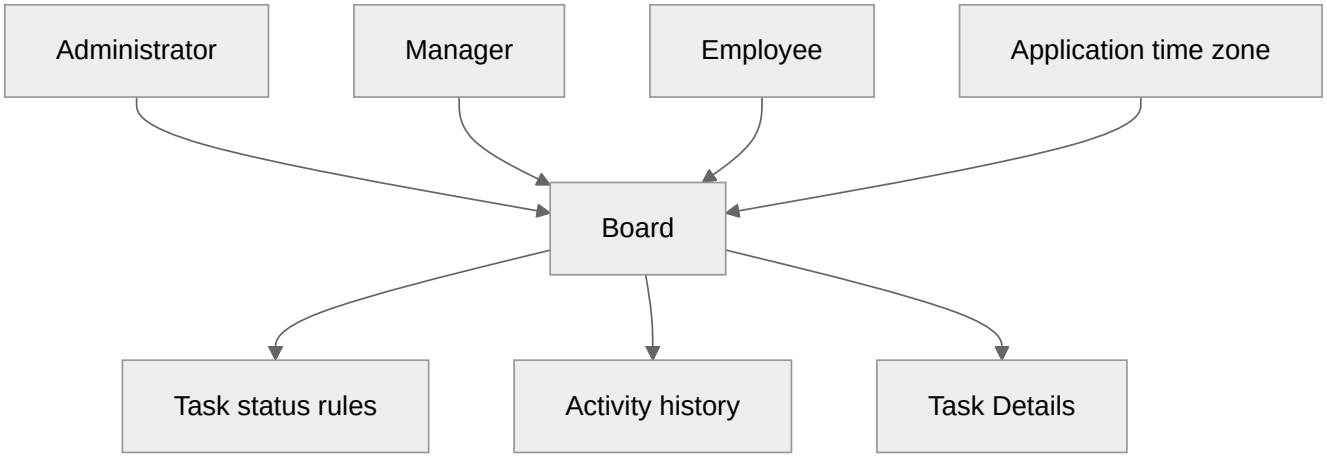
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2. System Overview

2.1 Module Map

Code	Module	Child modules / screens	Primary roles
BOARD	Board (TaskFlow Monitoring)	Board (Kanban)	Administrator, Manager, Employee, Custom role with `board.view`
TASK	Task Management (related)	Task scope, status transition table, activity history, Task Details	All users
ROLE	Role Configuration (related)	Grants `board.view`, `board.filters.use`, `tasks.status.change`	Administrator

2.2 Context Diagram



3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Administrator	Views, filters and moves any task	Organization (all tasks)
Manager	Views, filters and moves team tasks	Team, plus tasks the Manager created, is assigned or is notified about
Employee	Views, filters and moves own tasks	Own (assigned, created or notified)
Custom role	As granted by `board.view`, `board.filters.use`, `tasks.status.change`	Follows the user's task scope

3.2 Permission Matrix (functional view)

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Board › View board	`board.view`	Granted	Granted	Granted	Configurable

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Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Board › Use board filters (requires `board.view`)	`board.filters.use`	Granted	Granted	Granted	Configurable
Board › Update task status by drag (shared with Task Details)	`tasks.status.change`	Granted	Granted	Granted	Configurable
Task Management › View my tasks (cards in Own scope)	`tasks.view.own`	Always On	Always On	Always On	Always On
Task Management › View all tasks (Team / Organization cards, Assignee filter, menu location)	`tasks.view.all`	Granted	Granted	Denied	Configurable

4. Functional Requirements by Module

4.1 Module: Board (BOARD)

Purpose: Give every user a visual overview of their accessible tasks by status and the fastest way to move a task to its next allowed status. · **BRD reference:** BRD-Board §5 SCREEN 1, BR-BOARD-01...06, BR-TASK-03; global BR-06, BR-10, BR-11, BR-12, BR-17, BR-19.

4.1.1 Screen: Board (Kanban)

Field	Value
Route / location	`/board` – menu **TaskFlow Monitoring → Board** (with `tasks.view.all`) or **Task Management → Board** (everyone else)
Access	`board.view` (Administrator, Manager, Employee; Custom role as granted); content follows the user's task scope
Entry points	Menu item (shown only with `board.view`); **Board view** button on the Task List; Back from Task Details opened from a card

Screen elements

Element	Type	Description	Mandatory	Visible to
Header	Text	"N tasks across 5 stages" ("Loading board..." while loading)	–	`board.view`
Refresh	Button	Reloads the board	–	`board.view`
Project	Drop-down	Filter by project	No	`board.filters.use`
Assignee	Drop-down	Filter by assignee	No	`board.filters.use` and `tasks.view.all`
Status	Drop-down	Filter by status	No	`board.filters.use`
Deadline from, Deadline to	Date inputs	Inclusive deadline range; "to" cannot be set before "from"	No	`board.filters.use`
Clear	Button	Clears all filters; disabled while no filter is set	–	`board.filters.use`

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Element	Type	Description	Mandatory	Visible to
Columns	Lists	To Do, In Progress, In QA, Blocked, Completed, each with a count	–	`board.view`
Card	Draggable item	Task code, priority, title, project, assignee, deadline; overdue cards and deadlines highlighted	–	`board.view` (drag needs `tasks.status.change`)
Empty column	Text	"Drop tasks here"	–	`board.view`
Filtered-empty state	Text / button	"No tasks match the current filters." with **Clear filters**	–	`board.filters.use`

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-BOARD-01	The system shall show exactly the tasks of the user's task-list scope (Own, Team or Organization) in the columns To Do, In Progress, In QA, Blocked and Completed, with a count of cards in each column.	Must	`board.view`, `tasks.view.own`, `tasks.view.all`	REQ-BOARD-01, BR-BOARD-01, BR-06
FR-BOARD-02	The system shall not show cancelled or deleted tasks on the board.	Must	`board.view`	REQ-BOARD-02
FR-BOARD-03	The system shall show on each card the task code, priority, title, project, assignee and deadline; the column gives the status.	Must	`board.view`	REQ-BOARD-03
FR-BOARD-04	The system shall highlight a card and its deadline when the deadline has passed and the task is neither Completed nor Cancelled.	Must	`board.view`	REQ-BOARD-04, BR-BOARD-03, BR-12
FR-BOARD-05	The system shall order the cards in each column by deadline, soonest first, and ignore any attempt to re-order cards within a column.	Should	`board.view`	REQ-BOARD-05
FR-BOARD-06	The system shall show the total in the header as "N tasks across 5 stages", counting the cards shown.	Could	`board.view`	REQ-BOARD-06
FR-BOARD-07	The system shall let users with `tasks.status.change` drag a card to another column to change the task's status, and disable dragging for users without it.	Must	`tasks.status.change`	REQ-BOARD-07
FR-BOARD-08	The system shall check every move on the server against the permission, the user's access to the task and the transition table in BR-TASK-03, and refuse a disallowed move with "Cannot move a task from {from} to {to}.".	Must	`tasks.status.change`	REQ-BOARD-08, BR-BOARD-02, NFR-BOARD-02
FR-BOARD-09	The system shall change nothing when a card is dropped back into its own column.	Should	`tasks.status.change`	BR-BOARD-06

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ID	Requirement	Priority	Permission	BRD trace
FR-BOARD-10	The system shall move the card on screen as soon as it is dropped and, if the move is refused or fails, return it to its original column and show the server's reason or "Could not move {code}.".	Should	`tasks.status.change`	REQ-BOARD-09, NFR-BOARD-03, NFR-BOARD-04
FR-BOARD-11	The system shall confirm an accepted move with "{code} moved to {status}" (e.g. "TSK-000123 moved to In Progress").	Should	`tasks.status.change`	REQ-BOARD-10
FR-BOARD-12	The system shall record "Status changed from {X} to {Y}" in the task's history for every accepted move, plus "Task completed" when moved to Completed, and store the completion time.	Must	`tasks.status.change`	REQ-BOARD-11, BR-19
FR-BOARD-13	The system shall filter the loaded cards by Project, Status and an inclusive Deadline from/to range on screen without reloading, combining all chosen filters (AND).	Should	`board.filters.use`	REQ-BOARD-12, BR-BOARD-04, NFR-BOARD-01
FR-BOARD-14	The system shall offer the Assignee filter only to users with `tasks.view.all`.	Should	`tasks.view.all`	REQ-BOARD-13, BR-BOARD-05
FR-BOARD-15	The system shall show the filter row only to users with `board.filters.use`.	Should	`board.filters.use`	REQ-BOARD-14, BR-BOARD-05
FR-BOARD-16	The system shall clear every filter when **Clear** (filter row) or **Clear filters** (empty state) is clicked, and disable **Clear** while no filter is set.	Could	`board.filters.use`	REQ-BOARD-12, REQ-BOARD-15
FR-BOARD-17	The system shall show "Drop tasks here" in an empty column and "No tasks match the current filters." when the filters hide every loaded task.	Could	`board.view`	REQ-BOARD-15
FR-BOARD-18	The system shall reload the tasks and their latest statuses when the user clicks **Refresh**.	Must	`board.view`	REQ-BOARD-16
FR-BOARD-19	The system shall show "Could not load the board. Try again." when the board cannot be loaded.	Should	`board.view`	REQ-BOARD-01, REQ-BOARD-16
FR-BOARD-20	The system shall open Task Details when a card is clicked, with Back returning to the board.	Must	`tasks.view.own`	REQ-BOARD-17
FR-BOARD-21	The system shall show the Board menu item and open the screen only for users with `board.view`, and send everyone else back to their dashboard.	Must	`board.view`	REQ-BOARD-18
FR-BOARD-22	The system shall place the menu item under **TaskFlow Monitoring** for users with `tasks.view.all` and under **Task Management** for everyone else.	Could	`tasks.view.all`	REQ-BOARD-18

Use case: UC-BOARD-01 – View the board

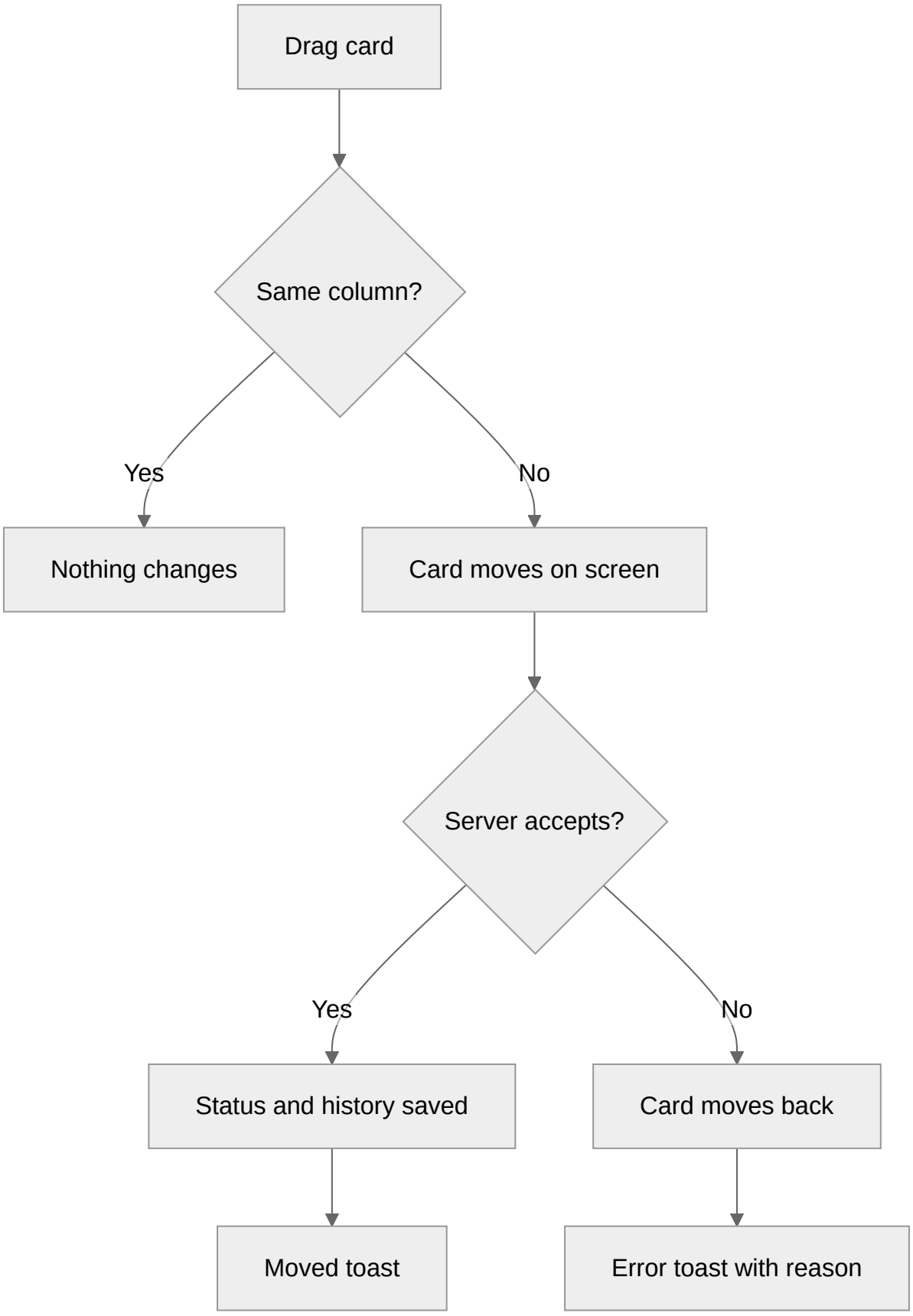
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Item	Detail
Actor	Employee, Manager or Administrator
Preconditions	Signed in; holds `board.view`
Trigger	Opens **Board** from the menu or **Board view** on the Task List
Main flow	1. System loads every task in the user's scope except cancelled and deleted tasks. 2. System places each card in its status column, ordered by deadline, with a count per column. 3. System highlights overdue cards. 4. System shows "N tasks across 5 stages".
Alternate flows	A1: **Refresh** – the board reloads with the latest statuses.
Exceptions	E1: Load failure → "Could not load the board. Try again." E2: No `board.view` → menu hidden; the screen sends the user back to the dashboard.
Postconditions	Nothing changed

Use case: UC-BOARD-02 – Move a card to a new status

Item	Detail
Actor	Employee, Manager or Administrator
Preconditions	Holds `board.view` and `tasks.status.change`; the task is in the user's scope
Trigger	Drags a card to another column
Main flow	1. Card moves on screen at once. 2. System checks the permission, task access and transition table. 3. System saves the status (and completion time when Completed) and records "Status changed from {X} to {Y}" (plus "Task completed"). 4. System shows "{code} moved to {status}".
Alternate flows	A1: Card dropped in its own column – nothing happens.
Exceptions	E1: Disallowed move (e.g. Completed → In Progress) → card returns; "Cannot move a task from Completed to InProgress." E2: Other failure → card returns; "Could not move {code}." E3: No `tasks.status.change` → cards cannot be dragged.
Postconditions	Accepted: new status saved and recorded, reminders stop for Completed. Refused: task unchanged on server and screen

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Use case: UC-BOARD-03 – Filter the board

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Item	Detail
Actor	User with `board.filters.use`
Preconditions	Board loaded
Trigger	Picks a Project, Assignee (view-all users), Status or Deadline range
Main flow	1. System narrows the loaded cards on screen, combining all filters (AND), with both deadline dates included. 2. Column counts and the header follow the cards shown. 3. **Clear** removes all filters.
Alternate flows	–
Exceptions	E1: No card matches → "No tasks match the current filters." with **Clear filters**.
Postconditions	Nothing changed on the server

Use case: UC-BOARD-04 – Open a task from a card

Item	Detail
Actor	Any board user
Preconditions	Board loaded
Trigger	Clicks a card
Main flow	1. System opens Task Details for the task. 2. User clicks Back. 3. System returns to the board.
Alternate flows	–
Exceptions	E1: Task no longer accessible → Task Details shows its fallback text.
Postconditions	Nothing changed

Field validations

Field	Rule	Error message
Deadline to	Cannot be set before Deadline from	– (the date picker blocks earlier dates)
Target column	Allowed by BR-TASK-03 from the card's current status	"Cannot move a task from {from} to {to}."

Business rules applied: BR-BOARD-01, BR-BOARD-02, BR-BOARD-03, BR-BOARD-04, BR-BOARD-05, BR-BOARD-06, BR-TASK-03; global BR-06, BR-10, BR-11, BR-12, BR-17, BR-19.

Notifications / side effects

Event	Who is notified	Channel
Card moved	Nobody – recorded in the task's activity history only	–
Card moved to Completed	Reminder engine sends no further reminders for the task	–

Messages

Situation	Message
Move accepted	"TSK-000123 moved to In Progress"
Move not allowed	"Cannot move a task from Completed to InProgress." (statuses vary)

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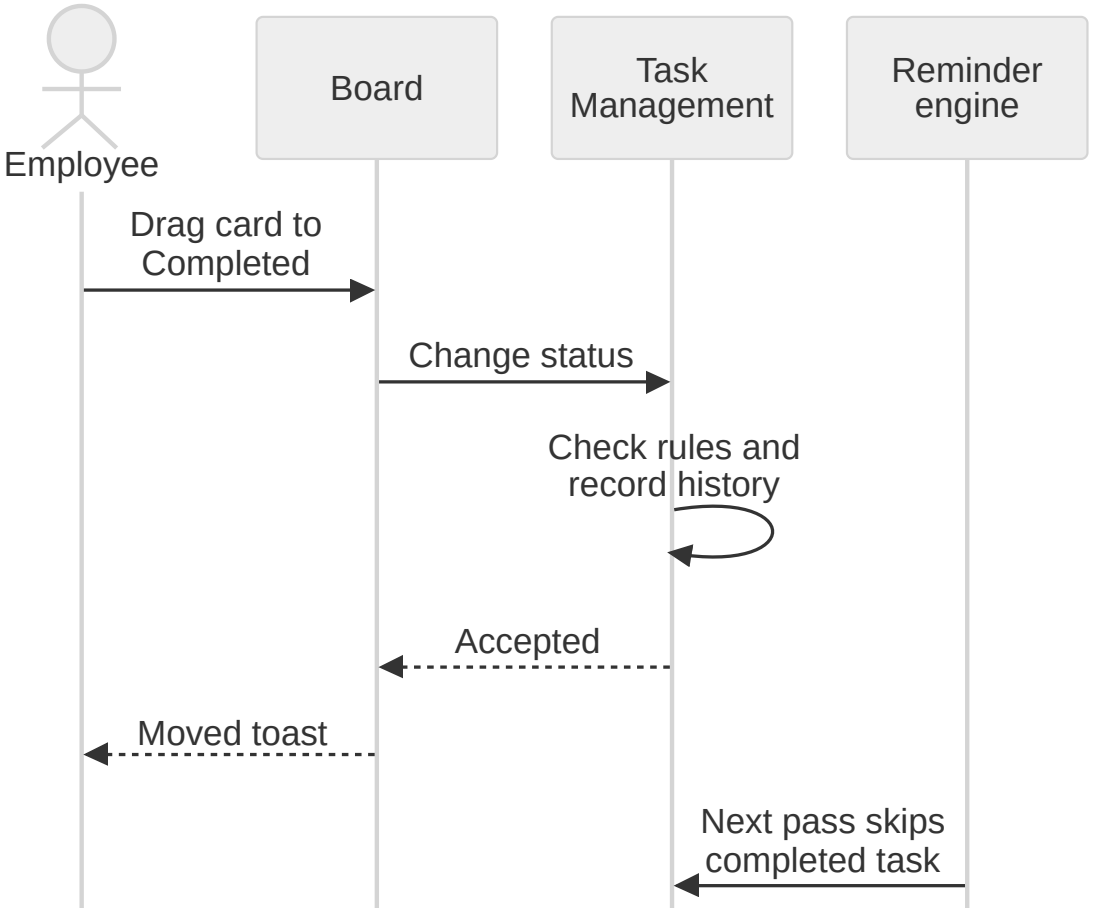
Situation	Message
Move failed (other)	"Could not move TSK-000123."
Load failed	"Could not load the board. Try again."
Filters hide all cards	"No tasks match the current filters."
Empty column	"Drop tasks here"
No permission	Menu item hidden; the screen sends the user back to their dashboard

5. Cross-Module Workflows

5.1 Move a task on the board → history → reminders

Step	Module	Actor / System	Action	FR reference
1	BOARD	Employee	Drags a card from In QA to Completed	FR-BOARD-07, FR-BOARD-10
2	TASK	System	Checks the transition table and the user's access, saves the status and completion time	FR-BOARD-08, FR-TASK-38, FR-TASK-40 (Task Management FRD)
3	TASK	System	Records "Status changed from InQA to Completed" and "Task completed"	FR-BOARD-12, FR-TASK-63
4	NOTIF	Reminder engine	Stops reminders for the task	FR-TASK-47
5	BOARD	System	Shows "TSK-... moved to Completed"	FR-BOARD-11

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6. Reports & Exports

Not applicable for this sub-module. Status reports are in `../../05-Reports/05.1-Reports/FRD-Reports.md`.

7. Notifications & Messages Catalogue

Not applicable for this sub-module. A status change on the board sends no notification or mail; it is recorded in the task's activity history. On-screen messages are listed in §4.1.1.

8. Data Requirements (functional view)

This sub-module stores no data of its own.

Data item	Description	Created by	Rules
Task (card)	Code, title, project, assignee, priority, deadline, status	Task Management	Cancelled and deleted tasks not shown; status updated by an accepted move
Overdue flag	Deadline passed and not Completed or Cancelled	System (on load)	Never stored

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Data item	Description	Created by	Rules
Activity history entry	"Status changed from X to Y", "Task completed"	System	Owned by Task Management; never changed by users

9. Non-Functional Requirements (functional impact)

ID	Requirement	Affects
NFR-BOARD-01	Filters apply on screen to the loaded tasks; one load per open or Refresh	Board (FR-BOARD-13, FR-BOARD-18)
NFR-BOARD-02	Status permission, task access and status rules checked on the server for every move	Board (FR-BOARD-08)
NFR-BOARD-03	Moves feel immediate and failures are visible	Board (FR-BOARD-10)
NFR-BOARD-04	The board never shows a status the server did not accept	Board (FR-BOARD-10)
NFR-BOARD-05	One transition table for the board and Task Details (BR-TASK-03)	Board, Task Details
NFR-BOARD-06	Drag-and-drop works in the supported browsers (global NFR-16)	Board
NFR-BOARD-07	Accessibility as the global NFRs (NFR-18)	Board

10. Assumptions & Open Issues

Type	Description	Owner
Assumption	The number of accessible tasks stays small enough to load the whole board at once (no paging)	Jojo George
Open issue	Large task volumes could slow the board because it is not paged (R-BOARD-01)	Jojo George
Open issue	`board.filters.use` is enforced only on the screen; filters only narrow data the user may already see (R-BOARD-02)	Jojo George
Open issue	Cancelled tasks are invisible on the board; use the Task List Status filter (R-BOARD-03)	Jojo George
Open issue	The Task List **Board view** button is shown without `board.view`; the board then sends the user back (R-BOARD-04)	Jojo George
Open issue	The deadline filter compares the deadline's calendar date in the user's browser time zone, while overdue uses the application time zone; the two can differ near midnight for users in another time zone	Jojo George

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11. Requirements Traceability Matrix

Test cases name the automated test files (`TaskFlow.UnitTests` / `TaskFlow.IntegrationTests`) whose names match the behaviour; coverage is inferred from the file names. "Manual UI" = checked by hand on the screen.

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-BOARD-01	FR-BOARD-01, FR-BOARD-19	Board	UC-BOARD-01	`TasksEndpointsTests.cs`, `TaskServiceTests.cs`
REQ-BOARD-02	FR-BOARD-02	Board	UC-BOARD-01	`TaskServiceTests.cs`
REQ-BOARD-03	FR-BOARD-03	Board	UC-BOARD-01	Manual UI
REQ-BOARD-04	FR-BOARD-04	Board	UC-BOARD-01	`TaskTrackingDomainTests.cs`
REQ-BOARD-05	FR-BOARD-05	Board	UC-BOARD-01	Manual UI
REQ-BOARD-06	FR-BOARD-06	Board	UC-BOARD-01	Manual UI
REQ-BOARD-07	FR-BOARD-07	Board	UC-BOARD-02	`CustomRolePermissionTests.cs`
REQ-BOARD-08	FR-BOARD-08	Board	UC-BOARD-02	`TaskTrackingDomainTests.cs`
REQ-BOARD-09	FR-BOARD-10	Board	UC-BOARD-02	Manual UI
REQ-BOARD-10	FR-BOARD-11	Board	UC-BOARD-02	Manual UI
REQ-BOARD-11	FR-BOARD-12	Board	UC-BOARD-02	`TaskManagementServiceTests.cs`
REQ-BOARD-12	FR-BOARD-13, FR-BOARD-16	Board	UC-BOARD-03	Manual UI
REQ-BOARD-13	FR-BOARD-14	Board	UC-BOARD-03	Manual UI
REQ-BOARD-14	FR-BOARD-15	Board	UC-BOARD-03	Manual UI
REQ-BOARD-15	FR-BOARD-16, FR-BOARD-17	Board	UC-BOARD-03	Manual UI
REQ-BOARD-16	FR-BOARD-18, FR-BOARD-19	Board	UC-BOARD-01	Manual UI
REQ-BOARD-17	FR-BOARD-20	Board	UC-BOARD-04	Manual UI
REQ-BOARD-18	FR-BOARD-21, FR-BOARD-22	Board	UC-BOARD-01	`ModuleBasedPermissionEndpointsTests.cs`

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BRD item	FRD requirement(s)	Screen	Use case	Test case
BR-BOARD-01	FR-BOARD-01	Board	UC-BOARD-01	`TaskServiceTests.cs`
BR-BOARD-02	FR-BOARD-08	Board	UC-BOARD-02	`TaskTrackingDomainTests.cs`
BR-BOARD-03	FR-BOARD-04	Board	UC-BOARD-01	`TaskTrackingDomainTests.cs`
BR-BOARD-04	FR-BOARD-13	Board	UC-BOARD-03	Manual UI
BR-BOARD-05	FR-BOARD-14, FR-BOARD-15	Board	UC-BOARD-03	Manual UI
BR-BOARD-06	FR-BOARD-09	Board	UC-BOARD-02	Manual UI

12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	Jojo George	Pending	–

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